

Tecnológico Nacional de México

Campus Saltillo

Ingeniería en Sistemas Computacionales

Computo en la nube

5.9 Despliegue de aplicación CRUD en la nube

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1. Creación de VM

Microsoft Azure Portal - Compute infrastructure | Virtual machines

Get started

Virtual machines

Name	Subscription	Resource Group	Location	Status	Operating system	Size	Public IP address	Disks	Update status
MV-Hugo	Azure for Students	CN-Hugo-Espinoza	Canada Central	Stopped (deallocat...	Linux	Standard_B2ats_v2	20.220.163.71	1	Enable periodic as...
srvapp	Azure for Students	CN-Hugo-Espinoza	Canada Central	Stopped (deallocat...	Linux	Standard_B2ats_v2	-	1	Enable periodic as...
srvdb	Azure for Students	CN-Hugo-Espinoza	Canada Central	Stopped (deallocat...	Linux	Standard_B2ats_v2	-	1	Enable periodic as...

Virtual machine: **srvapp**

Help me copy this VM in any region | Manage this VM with Azure CLI

Your VM has a default outbound IP, which is insecure and will no longer be assigned by default for new subnets after March 2026. To secure your VM and subnets and ensure future compatibility, follow guidance to add an explicit method of outbound and set your subnets to private.

Help me copy this VM in any region

Connect | Start | Restart | Stop | Hibernate | Capture | Delete | Refresh | Scale | Open in mobile

Essentials

Resource group (move): [CN-Hugo-Espinoza](#)

Status: Running

Location: Canada Central (Zone 1)

Subscription (move): [Azure for Students](#)

Subscription ID: fb79fd1c-41dc-40eb-97f9-2609429352c0

Availability zone: 1

Tags (edit) | Add tags

Operating system: Linux (ubuntu 24.04)

Size: Standard B2ats v2 (2 vcpus, 1 GiB memory)

Primary NIC public IP: -

Virtual network/subnet: [MV-Hugo-vnet/default](#)

DNS name: -

Health state: -

Time created: 4/30/2026, 3:29 AM UTC

Virtual machine: **srvdb**

Help me copy this VM in any region | Manage this VM with Azure CLI

Your VM has a default outbound IP, which is insecure and will no longer be assigned by default for new subnets after March 2026. To secure your VM and subnets and ensure future compatibility, follow guidance to add an explicit method of outbound and set your subnets to private.

Help me copy this VM in any region

Connect | Start | Restart | Stop | Hibernate | Capture | Delete | Refresh | Scale | Open in mobile | Feedback | CLI / PS

Essentials

Resource group (move): [CN-Hugo-Espinoza](#)

Status: Running

Location: Canada Central (Zone 1)

Subscription (move): [Azure for Students](#)

Subscription ID: fb79fd1c-41dc-40eb-97f9-2609429352c0

Availability zone: 1

Tags (edit) | Add tags

Operating system: Linux

Size: Standard B2ats v2 (2 vcpus, 1 GiB memory)

Primary NIC public IP: -

Virtual network/subnet: [MV-Hugo-vnet/default](#)

DNS name: -

Health state: -

Time created: 4/30/2026, 2:38 AM UTC



2. Evidencia

< > ↻ Not secure 20.220.163.71/usuarios/controladores/controlador.php

Mostrando Registros de Usuarios

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Show entries Buscar

eMail	Fecha	Estatus
Lo sentimos. No se encontraron registros.		

No hay registros aún.

[Anterior](#) [Siguiente](#)

< > ↻ Not secure 20.220.163.71/usuarios/controladores/controlador.php

Agregando Usuario

[Salir](#)

Correo Electrónico

Contraseña

Estatus

[Guardar](#)

< > ↻ Not secure 20.220.163.71/usuarios/controladores/controlador.php

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Show entries Buscar

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chuy@gmail.com	2026-04-30	A

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[Anterior](#) [1](#) [Siguiente](#)

3. Conclusiones

Durante esta práctica se logró desplegar una aplicación CRUD en la nube utilizando tres máquinas virtuales en Azure. Se configuró la comunicación entre la capa de presentación (srvrep), la capa de aplicación (srvapp) y la base de datos (srvdb), permitiendo realizar operaciones de inserción y consulta correctamente. Esto permitió comprender el funcionamiento de una arquitectura de tres capas y el proceso de implementación de aplicaciones en entornos cloud.